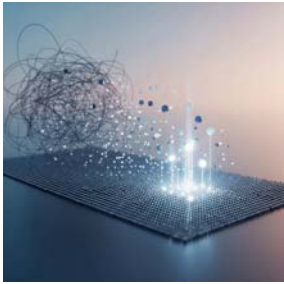


A4.2.1 Describe the significance of data cleaning.



A4.2.1 Data cleaning

Before a machine learning model can learn useful patterns, the data it trains on must be accurate, consistent, and relevant. Raw, real-world data is almost never in that condition. Data cleaning is the process of detecting and correcting problems in a dataset before training begins. This page covers why data quality matters, the six categories of data problems and the techniques for addressing each, and the preprocessing steps of normalisation and standardisation.

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Key Questions

- What is the 'Garbage In, Garbage Out' principle and why does it matter for machine learning?
- What are the six categories of data problem covered in this topic, and what is the consequence of leaving each unaddressed?
- What is the difference between an outlier and incorrect data, and why does that distinction affect how you handle each?
- What are the three techniques for handling missing data, and what are the trade-offs between them?
- What is the difference between normalisation and standardisation, and when would each be applied?

IB Command Terms

Describe

Give a detailed account of the features or characteristics. For data cleaning, this means: name the technique, explain what problem it solves, describe how it works, and explain why it matters for model performance.

What to include

For each data problem: the problem definition, a concrete example, the consequence of ignoring it, and the technique(s) used to address it. For normalisation and standardisation: the definition, the formula or transformation, and the reason it is needed.

The Impact of Data Quality on Model Performance

A machine learning model learns exclusively from the data it is given. It has no external reference point, no ability to question whether the data is accurate, and no way to compensate for systematic errors. This makes data quality the single most important factor in determining whether a trained model will be useful.

The principle is known as **Garbage In, Garbage Out (GIGO)**. Regardless of how sophisticated the model architecture is, training on poor-quality data produces a poor-quality model. The model learns the patterns in the data it receives and if those patterns reflect errors, gaps, or biases, the model encodes those problems into its predictions.

INACCURATE PREDICTIONS

If a model is trained on incorrect data, for example, incorrectly labelled medical images, it learns false patterns. Its predictions on new data will reflect those false patterns, potentially producing dangerous outcomes.

BIASED OUTCOMES

If the training data reflects historical biases, for example, a hiring dataset in which certain groups were systematically underrepresented, the model will learn and often amplify those biases. Discriminatory decisions follow.

POOR GENERALISATION

A model trained on noisy or irrelevant data may learn patterns that exist only in the training set and not in the real world. When presented with new data, it performs poorly because it has fitted to noise rather than to meaningful structure.

Why data cleaning takes most of the time

In practice, data scientists typically spend 60 to 80 per cent of a project's time on data cleaning and preprocessing rather than on building or tuning models. Real-world data arrives from multiple sources with inconsistent formats, missing values, and errors introduced by human input. A model trained without addressing these problems will perform poorly regardless of how carefully the model architecture is designed.

Techniques for Handling Common Data Problems

The IB syllabus identifies six categories of data problem. Each requires a different approach. The distinction between some categories, particularly outliers and incorrect data, is important and may be tested in the examinations. Click on the headings to see the details.

1. Missing Data

The problem could be that one or more values are absent from a record. A patient record with no blood pressure reading, an online form where the user left the age field blank, or a sensor that failed to log a reading.

Consequence if ignored: Many ML algorithms cannot process null values and will crash or produce errors. If rows with missing values are simply dropped without consideration, the remaining dataset may be non-representative.

The data could be missing for a number of reasons:

Missing Completely At Random (MCAR): the absence is unrelated to any other variable.

Missing At Random (MAR): the absence is related to another observed variable.

Missing Not At Random (MNAR): the absence is related to the missing value itself - for example, patients with very high blood pressure may be less likely to report it.

The problem: One or more values are absent from a record. A patient record with no blood pressure reading, an online form where the user left the age field blank, or a sensor that failed to log a reading.

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Why data may be missing: Missing Completely At Random (MCAR): the absence is unrelated to any other variable. Missing At Random (MAR): the absence is related to another observed variable. Missing Not At Random (MNAR): the absence is related to the missing value itself - for example, patients with very high blood pressure may be less likely to report it.

TECHNIQUE 1: DELETION

Remove any row containing a missing value. Simple and produces a complete dataset.

TECHNIQUE 2: IMPUTATION

Replace the missing value with a calculated substitute. Common strategies: mean

TECHNIQUE 3: PREDICTIVE MODELLING

Train a separate ML model to predict the missing value

However, if many rows have at least one missing value, a large proportion of the data may be lost. Risk: if missing values are not random, deletion introduces bias.

(arithmetic average of all non-missing values in that column), median (middle value when sorted, more robust to outliers), or mode (most frequent value, used for categorical data). Preserves all rows but introduces artificial values.

using the other features of that row as inputs. For example, predict missing salary from job title, years of experience, and location. More accurate than mean imputation but computationally expensive and adds complexity to the pipeline.

2. Outliers

The problem: A data point that is statistically distant from the rest of the data. An age recorded as 210, a house price of 5 (in a dataset where prices range from 100,000 to 2,000,000), or a transaction of 0.001 seconds in a dataset where the minimum is typically 30 seconds.

Important distinction: An outlier is not the same as incorrect data. An outlier may be a genuine extreme value such as a billionaire's salary in a dataset of employee incomes is a true outlier, not an error. Whether to treat a value as an error or a legitimate extreme requires domain knowledge and investigation.

Consequence if ignored: A single extreme outlier can disproportionately shift the model's learned parameters. In a regression model, one extreme value can pull the fitted line away from where most of the data lies, degrading predictions for the majority of cases.

Technique 1: Investigate first: Determine whether the outlier is an error or a legitimate value. Check against source data, apply domain knowledge, or consult a domain expert.

Technique 2: Remove: If confirmed as an error with no correct value available, remove the row. If a legitimate value that will distort the model, removal may still be appropriate with documentation.

Technique 3: Cap (Winsorisation): Replace values beyond a defined threshold with the threshold value. For example, cap all ages above 100 at 100. Preserves the row while limiting the influence of the extreme value.

Technique 4: Log transformation: Apply a logarithmic transformation to a skewed feature, compressing the range of extreme values. Useful when a feature spans several orders of magnitude.

3. Incorrect Data

The problem: Values that are factually wrong, not merely statistically unusual. A date of birth entered as 2045 (in the future), a salary entered as -50000 (negative), a country code entered as 'XZ' (non-existent), or a label misapplied by a human annotator. Incorrect data may or may not be detectable as an outlier.

Distinction from outliers: An outlier is unusual relative to the dataset's distribution. Incorrect data is wrong relative to the real world. A salary of 250,000 may not be an outlier in a senior executive dataset, but if that row belongs to an intern, it is incorrect data. Outlier detection algorithms will not catch this; domain validation rules are required.

Consequence if ignored: The model learns false relationships. A classification model trained on mislabelled images learns the wrong associations between features and classes, this is particularly damaging because the model cannot distinguish labelling errors from true patterns.

Technique 1: Validation rules: Apply constraints that values must satisfy: age must be between 0 and 120; salary must be positive; country code must exist in a reference list. Flag or reject values that violate these rules.

Technique 2: Cross-checking: Verify values against an external reference dataset or a related column. If a postcode does not match the stated city, one of the two values is likely incorrect.

Technique 3: Correction or removal: If the correct value can be determined, correct it. If not, treat the field as missing and apply a missing data technique.

4. Duplicate Data

The problem: The same record appears more than once in the dataset. This may be exact duplication (identical rows) or near-duplication (same entity with minor variations, such as 'J. Smith' and 'John Smith' at the same address).

Consequence if ignored: During training, the model encounters the duplicated record more frequently than other records. It assigns more weight to the patterns in that record than its true frequency in the real world warrants. For classification, this can cause the model to overfit to the characteristics of the duplicated examples.

Technique: Remove duplicates: Identify rows that are identical across all columns and remove all but one instance. For near-duplicates, use matching algorithms (comparing names, addresses, and other identifiers) to identify likely duplicates and consolidate them into a single canonical record.

5. Irrelevant Data

The problem: Features (columns) that have no logical relationship to the target variable. A customer's telephone number when predicting purchase likelihood. A unique ID when predicting house prices. A timestamp column when the time of data entry is unrelated to the outcome.

Consequence if ignored: Irrelevant features add noise. With enough irrelevant features, a model may find spurious correlations, patterns that exist in the training data by chance but do not generalise. This is particularly problematic when the dataset has many features relative to the number of examples. Irrelevant features also increase training time and memory use.

Technique: Feature selection: Identify and remove features that do not contribute predictive value. Methods include: removing features with near-zero variance; removing features that are logically unrelated to the prediction task; and statistical tests that measure the correlation between each feature and the target variable.

6. Improperly Formatted Data

The problem: The same real-world value is represented in multiple inconsistent formats. A country column containing 'USA', 'U.S.A.', 'United States', and 'america'. A date column with entries in both DD/MM/YYYY and MM/DD/YYYY formats. A text column with inconsistent capitalisation.

Consequence if ignored: The model treats each unique string as a different category. 'USA', 'U.S.A.', and 'United States' become three separate values with separate learned associations, splitting what should be one feature into several weak fragments. Patterns that should generalise across all US records are diluted.

Technique: Format standardisation: Transform all representations of the same value to a single agreed format. Convert all country names to ISO codes; convert all dates to ISO 8601 (YYYY-MM-DD); apply consistent capitalisation (all lower case or title case). This must be applied before any other processing that depends on these values.

Note on terminology: This is referred to in the syllabus as transforming improperly formatted data. The word 'standardisation' here means making formats consistent, it is distinct from statistical standardisation (rescaling numerical data to mean 0, standard deviation 1), which is covered in the next section.

Normalization and Standardization: Crucial Preprocessing Steps

After the six cleaning steps above, numerical features in a dataset may still vary enormously in scale. A salary column might range from 20,000 to 500,000; an age column from 18 to 90. Many ML algorithms are sensitive to this difference in scale, they may incorrectly infer that salary is a more important predictor than age simply because its values are numerically larger. Normalisation and standardisation address this by rescaling all features to a comparable range.

A model is predicting whether a loan will be repaid. The dataset has two features: annual income (ranging from 15,000 to 250,000) and credit score (ranging from 300 to 850). Which feature do you think a distance-based algorithm might incorrectly treat as more important, and why?

NORMALISATION (MIN-MAX SCALING)

Rescales each feature to a fixed range, typically [0, 1]. The minimum value in the column becomes 0; the maximum becomes 1; all other values are scaled proportionally between them.

Formula: $x' = (x - x_{\min}) / (x_{\max} - x_{\min})$

Example: A salary of 55,000 in a column where min = 20,000 and max = 200,000.
Normalised value = $(55,000 - 20,000) /$

STANDARDISATION (Z-SCORE SCALING)

Rescales each feature so that it has a mean (average) of 0 and a standard deviation of 1. Standard deviation is a measure of how spread out the values are: a low standard deviation means values cluster near the mean; a high one means they are spread widely.

Formula: $x' = (x - \text{mean}) / \text{standard deviation}$

Example: A salary of 55,000 where mean = 80,000 and standard deviation = 30,000. Standardised value = $(55,000 - 80,000) / 30,000 = -25,000 / 30,000 = -0.83$ (below average by 0.83 standard deviations).

$(200,000 - 20,000) = 35,000 / 180,000 = 0.194$

Best used when: the feature has a known, bounded range and outliers are not a concern. The output range is predictable and interpretable.

Best used when: the feature does not have a known bounded range, or when outliers are present (standardisation is more robust to outliers than normalisation).

COMMON EXAM MISTAKE

The word 'standardisation' appears in two different contexts on this page. Format standardisation means making inconsistent text representations consistent (all country names in the same format). Statistical standardisation (z-score scaling) means rescaling numerical data to mean 0 and standard deviation 1. These are completely different operations. In an exam answer, make clear which meaning you are using.

Worked example: cleaning a single record

The table below shows one row from a raw customer dataset, the problems identified in each field, and the result after each cleaning step is applied. Row colour indicates the type of problem.

ROW 1 = RAW DATA

ROW 2 = PROBLEM IDENTIFIED

ROW 3 = AFTER CLEANING

Name	Age	Salary	Country	Purchase?	Customer_ID
alice smith	210	55000	U.S.A.	(empty)	C-00142
--	<i>Outlier: age 210 is impossible</i>	--	<i>Format inconsistency</i>	<i>Missing value</i>	<i>Irrelevant feature</i>

Alice Smith (format corrected)	(Row removed, outlier)	55000 (no change needed)	USA (format standardised)	0.52 (imputed (column mean))	(Column removed, irrelevant)
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READING THE TABLE

Name ('alice smith'): improperly formatted - inconsistent capitalisation. Corrected to 'Alice Smith'.

Age (210): outlier - investigated and confirmed as a data entry error (should be 21). The row is removed here; alternatively the age could be corrected if the correct value can be verified.

Salary (55000): no problem identified - value is within a plausible range.

Country ('U.S.A.'): improperly formatted - non-standard representation. Standardised to 'USA'.

Purchase? (empty): missing value - imputed using the column mean (0.52, representing the proportion of customers who made a purchase).

Customer_ID ('C-00142'): irrelevant feature - a unique identifier has no predictive value and is removed.

Summary: six problems and their techniques

Use this table for revision. For each problem, you should be able to state the definition, give an example, explain the consequence, and describe at least one technique.

PROBLEM TYPE	DESCRIPTION	CONSEQUENCE IF IGNORED	TECHNIQUE(S)
Missing data	Empty cells where a value should exist	Model errors or crashes on null values; biased results if missing data is not random	<i>Deletion (remove row); mean/median imputation; predictive modelling</i>
Outliers	Values statistically distant from the rest of the data	Model overweights the extreme value; predictions skewed toward rare cases	<i>Investigate first; then: remove, cap (Winsorise), or apply log transformation</i>

PROBLEM TYPE	DESCRIPTION	CONSEQUENCE IF IGNORED	TECHNIQUE(S)
Incorrect data	Values that are factually wrong - not just unusual	Model learns false patterns from the errors	<i>Validation rules; cross-checking against reference data; correction or removal</i>
Duplicate data	Identical or near-identical records appearing more than once	Duplicated records are seen more often during training, causing the model to overfit to those patterns	<i>Identify and remove; or consolidate near-duplicates into one canonical record</i>
Irrelevant data	Features with no logical connection to the target variable	Adds noise; may cause the model to find spurious correlations; increases training time	<i>Feature selection: remove columns that do not contribute predictive value</i>
Improperly formatted data	Inconsistent representation of the same value (USA / U.S.A. / United States)	The model treats different representations as different categories, fragmenting useful information	<i>Format standardisation: transform all representations to one consistent format</i>

Flashcards

Use this Quizlet to learn all the key terms in this section.



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode 

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

The principle that states the quality of a model's output is determined by the quality of its input data is known as:

A. ☐ Moore's Law

B. ☒ "Garbage In, Garbage Out" (GIGO)

C. ☐ The Turing Test

D. ☐ The "No Free Lunch" Theorem

2

A dataset of employee information has a 'Salary' column, but several entries are empty. Filling these empty cells with the average salary of all other employees is an example of:

A. ☐ Normalization

B. ☐ Deletion

C. ☒ Imputation

D. ☐ An outlier

3

In a dataset of student records, one student's age is listed as 210. This data point is best described as:

A. ☐ Missing data

B. ☐ A duplicate

C. ☐ Irrelevant data

D. ☒ An outlier

4

A 'City' column contains the entries "New York", "new york", and "NY". The process of changing all of these to a single format like "New York" is called:

A. ☒ Standardization of data format

- B. ☐ Imputation
- C. ☐ Deletion
- D. ☐ Normalization

5

What is the primary purpose of Normalization and Standardization?

- A. ☐ To fill in missing values.
- B. ☒ To rescale numerical features to a common scale to prevent bias.
- C. ☐ To remove duplicate rows from the dataset.
- D. ☐ To filter out irrelevant columns.

6

A data scientist is building a model to predict house prices. The dataset contains a column for each homeowner's unique ID number. This column is an example of:

- A. ☐ Missing data
- B. ☐ Duplicate data
- C. ☐ An outlier
- D. ☒ Irrelevant data

7

Which technique for handling missing data results in the loss of other potentially useful information in a record?

- A. ☐ Imputation with the median

B. ☐ Predictive modelling

C. ☒ Deletion

D. ☐ Imputation with the mean

8

Normalization typically rescales data to which range?

A. ☐ A mean of 0 and a standard deviation of 1

B. ☐ -1 to 1

C. ☐ 0 to 100

D. ☒ 0 to 1

9

The process of identifying and correcting errors, inconsistencies, and inaccuracies in a dataset is known as:

A. ☐ Feature selection

B. ☐ Inference

C. ☐ Model training

D. ☒ Data cleaning

10

A dataset contains two identical rows for the same customer. This is an example of:

A. ☐ Missing data

B. ☐ Irrelevant data

C. ☒ Duplicate data

D. ☐ An outlier

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